Partial rebalancing of a delta hedge under fixed and proportional transaction costs: a three-phase solution

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Abstract

A short option position hedged with the underlying asset carries, between rebalancing dates, a hedge error: the difference between the model delta and the share holding. Correcting it incurs a fixed cost per trade and a cost proportional to the traded value. We determine the cost-optimal size of the correction in a one-period model. To leading order in the horizon, the P&L attributable to the hedge error is linear in the error; any nonlinearity in the value of a correction therefore originates in the risk functional used to evaluate that P&L. We prove a general criterion: for a convex risk functional ρ, an interior (partial) correction is optimal only if the proportional cost per unit traded lies in the interval $(\rho'(0^+), \rho'(u))$, where u is the current hedge error. A risk functional linear in the P&L standard deviation gives an empty interval and hence only full or zero corrections; a functional quadratic in it gives an interval with lower endpoint zero. For a functional containing both terms, the optimal correction is obtained in closed form. The parameter space divides into three regions (full, partial, none) whose boundaries are given explicitly and meet at a triple point. The relative width of the partial region equals $\gamma u S \sigma \sqrt{h}/\lambda$, the ratio of the quadratic to the linear coefficient scaled by the error. We also analyse an earlier exponential risk functional that produced a partial optimum only for proportional costs near 24%, and identify the error in it. A Monte Carlo simulation of the 20-week call example of Hull compares the rule with full rebalancing on a fixed schedule.

1 Introduction

Hull's textbook treats the delta hedge of 10^5 short call options with shares of the underlying, rebalanced weekly [1, ch. 19]. Between rebalancing dates the delta changes and the share holding is not equal to the model delta. In the absence of transaction costs the hedge would be restored completely at each date. With a fixed cost per trade and a proportional cost per unit of traded value, the size of the correction is a decision variable: given a hedge error of u shares, how many shares $x \in [0, u]$ should be traded?

This note originated in a numerical calculation. An exponential risk functional, combined with the two costs, gave an optimum strictly inside $(0, u)$, but only for proportional costs within about one percentage point of 24%. Section 4 shows that this functional does not represent the risk of the position. The remaining sections address the question: *which property of the risk functional determines whether a partial correction can be optimal?*

Results:

1. Lemma 2: for convex ρ, a partial correction is optimal only if the proportional cost per unit traded, bS, satisfies $\rho'(0^+) < bS < \rho'(u)$.

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2. The hedge-error P&L is linear in the error (Lemma 1). A constraint on the P&L standard deviation contributes a term linear in the error; a variance penalty contributes a quadratic term (Section 5).

3. For the combined functional the problem is solved exactly. The parameter space has three regions with explicit boundaries and a triple point (Theorem 1, Proposition 1). The partial region has relative width $\gamma u S\sigma\sqrt{h}/\lambda$ and vanishes as $\gamma \to 0$.

4. A simulation of the complete 20-week hedge applies the rule at every rebalancing date (Section 8).

After this work was completed, the author became aware that no-transaction regions of a related kind appear in the literature on option hedging with transaction costs, for example [2], where they are derived for a multi-period problem with proportional costs in the limit of small costs. The present note treats a single decision with fixed and proportional costs and solves it exactly.

2 Set-up

Position. The position is short n European calls with value $V(t,S)$, long H shares, and a bank account with balance B ($B<0$ denotes borrowing) that accrues interest at the risk-free rate r, $\mathrm{d}B = rB\,\mathrm{d}t$. At inception, after the calls are sold and before any shares are bought, $B = nV_0$. The share price follows geometric Brownian motion, $\mathrm{d}S = \mu S\,\mathrm{d}t + \sigma S\,\mathrm{d}W$. With delta $\Delta = \partial V/\partial S$, the share holding that cancels the first-order exposure is $n\Delta$. The hedge error is

$$g = n\Delta - H, \tag{1}$$

and in per-option units $g/n = \Delta - \Lambda$, where $\Lambda = H/n \in [0,1]$ is the hedge ratio. All quantities are real-valued; fractional shares are allowed.

P&L over a horizon h. The error is assumed to persist for a time h, for example until the next rebalancing date.

Lemma 1 (Linearity in the hedge error)**.** *Over $[t,t+h]$ the change in value of $\Pi = -nV+HS+B$ is*

$$\delta\Pi = -g\,(\delta S - rSh) + A + o(h), \tag{2}$$

where A does not depend on H. To leading order in h, the term depending on g is Gaussian with mean $-g(\mu-r)Sh$ and standard deviation $|g|\,S\sigma\sqrt{h}$.

Proof. By Itô's formula, $\delta V = \Delta\,\delta S + \frac{1}{2}\Gamma(\delta S)^2 + \Theta h + o(h)$. Let $\tilde{B} = B + HS$ be the balance with the share position converted to cash; for a given total value of the position, $\tilde{B}$ does not depend on H. The account accrues $rBh = r\tilde{B}h - rHSh$. Then

$$\delta\Pi = -(n\Delta - H)\,\delta S - rHSh - n\big(\tfrac{1}{2}\Gamma(\delta S)^2 + \Theta h\big) + r\tilde{B}h + o(h).$$

Adding and subtracting $rn\Delta Sh$ gives the stated form with $A = -n(\frac{1}{2}\Gamma(\delta S)^2+\Theta h)+r\tilde{B}h-rn\Delta Sh$, independent of H. With $\delta S = S(\mu h + \sigma\sqrt{h}\,Z)$, $Z\sim\mathcal{N}(0,1)$, the g-term has the stated mean and standard deviation. Its covariance with A involves $\mathbb{E}[(\delta S - \mathbb{E}\delta S)^3] = O(h^2)$ and $\mathbb{E}[\delta S]\,\mathrm{Var}[\delta S] = O(h^2)$, both $o(h)$. □

The g-dependent part of $\delta\Pi$ is linear in g. The position therefore supplies no curvature in g; any dependence of the marginal value of a correction on the size of the remaining error must come from the risk functional applied to $\delta\Pi$.

The drift term $-g(\mu-r)Sh$ is a directional exposure and is excluded from the hedging objective, which is equivalent to setting $\mu = r$ in the evaluation of the error. Retaining it shifts the target holding by a constant and does not change the structure of the results.

Decision problem. Let $u = |g| > 0$ and, without loss of generality, $g > 0$. Buying $x \in [0, u]$ shares reduces the error to $u - x$. Trades with $x > u$ or in the opposite direction are dominated: for $x > u$, the trade $2u - x$ leaves an error of equal magnitude at lower cost. A trade of $x > 0$ shares costs

$$c(x) = a + bSx, \qquad c(0) = 0, \tag{3}$$

with fixed cost $a \geq 0$ and proportional cost $b \geq 0$. Let $\rho(g) \geq 0$, $\rho(0) = 0$, be the risk functional, in dollars, assigned to an error g held over the horizon. The net benefit of trading x shares is

$$\mathcal{N}(x) = \rho(u) - \rho(u - x) - c(x), \qquad \mathcal{N}(0) = 0, \tag{4}$$

and the problem is $x^* = \arg\max_{x \in [0,u]} \mathcal{N}(x)$.

3 A necessary condition for partial correction

Let $\beta = bS$ denote the marginal transaction cost per share. On $(0, u]$,

$$\mathcal{N}'(x) = \rho'(u - x) - \beta. \tag{5}$$

The argument of ρ' is the *remaining* error $u - x$. At the start of the trade ($x \to 0^+$) the remaining error is u; at the end ($x = u$) it is 0. The slope of $\mathcal{N}$ at the two ends is therefore

$$\mathcal{N}'(0^+) = \rho'(u) - \beta, \qquad \mathcal{N}'(u) = \rho'(0^+) - \beta. \tag{6}$$

If ρ is convex, $\rho'(u - x)$ is non-increasing in x, so $\mathcal{N}'$ decreases from $\mathcal{N}'(0^+)$ to $\mathcal{N}'(u)$. Since c jumps by a at $x = 0$, $\lim_{x \to 0^+} \mathcal{N}(x) = -a$.

Lemma 2 (Marginal-cost interval)**.** *Let ρ be convex on $[0, u]$ and differentiable on $(0, u]$, with right derivative $\rho'(0^+)$ at 0.*

1. *If $\mathcal{N}'(u) \geq 0$, i.e. $\beta \leq \rho'(0^+)$: $\mathcal{N}$ is non-decreasing on $(0, u]$. The candidates are $x = u$ and $x = 0$; $x^* = u$ (*full*) if $\mathcal{N}(u) > 0$, otherwise $x^* = 0$.*
2. *If $\mathcal{N}'(0^+) > 0 > \mathcal{N}'(u)$, i.e. $\rho'(0^+) < \beta < \rho'(u)$: $\mathcal{N}$ increases and then decreases on $(0, u]$. The candidates are $\hat{x} \in (0, u)$ with $\mathcal{N}'(\hat{x}) = 0$, i.e. $\rho'(u - \hat{x}) = \beta$ (unique if ρ is strictly convex), and $x = 0$; $x^* = \hat{x}$ (*partial*) if $\mathcal{N}(\hat{x}) > 0$, otherwise $x^* = 0$.*
3. *If $\mathcal{N}'(0^+) \leq 0$, i.e. $\beta \geq \rho'(u)$: $\mathcal{N}$ is non-increasing on $(0, u]$ and bounded above by $-a \leq 0$, so $x^* = 0$ (*none*).*

Proof. Case 1: $\mathcal{N}'(x) \geq \mathcal{N}'(u) \geq 0$ on $(0, u]$, so $\max_{(0,u]} \mathcal{N} = \mathcal{N}(u)$. Case 3: $\mathcal{N}'(x) \leq \mathcal{N}'(0^+) \leq 0$ on $(0, u]$, so $\sup_{(0,u]} \mathcal{N} = \lim_{x \to 0^+} \mathcal{N} = -a \leq 0 = \mathcal{N}(0)$. Case 2: $\mathcal{N}'$ is non-increasing, positive near 0 and negative near u, so $\max_{(0,u]} \mathcal{N}$ is attained where $\mathcal{N}'$ changes sign. In each case the maximum over $(0, u]$ is compared with $\mathcal{N}(0) = 0$. □

Figure 1 shows the three cases for the risk functional of Section 5. In terms of marginal quantities: $\rho'(u)$ is the marginal risk reduction per share at the start of the trade, $\rho'(0^+)$ the marginal risk reduction as the error reaches zero, and β the marginal cost. An interior optimum exists only if the marginal cost lies strictly between these two values. The length of this interval is

$$\rho'(u) - \rho'(0^+) = \int_0^u \rho''(g)\, \mathrm{d}g. \tag{7}$$

Two consequences:

- if ρ is linear in $|g|$, the interval is empty and $x^* \in \{0, u\}$;
- if $\rho'(0^+) = 0$, the interval is $(0, \rho'(u))$, so for every $0 < b < \rho'(u)/S$ any trade made is partial and $x^* = u$ never occurs.

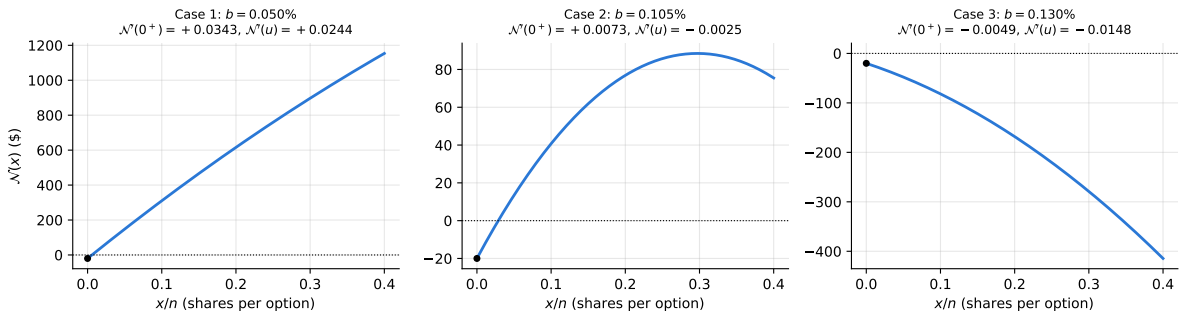


Figure 1: Net benefit $\mathcal{N}(x)$ in the three cases of Lemma 2, for the functional and parameters of Sections 5 and 7. The dot marks $\lim_{x\to 0^+} \mathcal{N} = -a$; the titles give the end slopes (6).

4 An exponential risk functional

The calculation that motivated this note used

$$\rho_{\exp}(g) = S \exp\Big(g\mu\tau - \tfrac{1}{2}g^2\sigma^2\tau + g\sigma\sqrt{\tau}\Big), \tag{8}$$

with τ the time to expiry, g in shares per option, and transaction costs compounded to expiry by $e^{r\tau}$. For $S = 49$, $\sigma = 0.2$, $\mu = r = 0.05$, $\tau = 1$, $u = 0.4$ per option and $a = 0.02$, $\rho_{\exp}$ is convex on $[0, u]$ and Lemma 2 gives the interval

$$23.8\% < b < 24.5\%,$$

of length 0.7 percentage points, at b of order $\sigma\sqrt{\tau} = 20\%$ (Figure 2). For all smaller b the optimum is $x^* = u$.

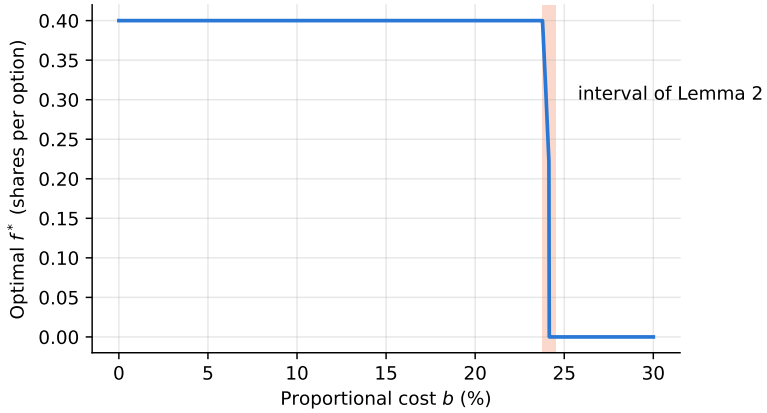


Figure 2: Exponential risk functional: optimal correction as a function of the proportional cost b. The shaded interval is that of Lemma 2.

The functional is not a risk measure of the hedge error:

- It has the form of the $+1\sigma$ terminal value of a portfolio continuously rebalanced to hold a constant fraction g of its wealth in the stock, for which $\log(\text{wealth/initial}) = (r + g(\mu - r))\tau - \frac{1}{2}g^2\sigma^2\tau + g\sigma W_\tau$. The hedge error is a static share position whose P&L is linear in g (Lemma 1); the curvature of $\rho_{\exp}$ in g belongs to the constant-fraction portfolio, not to the hedge error.
- g is a number of shares per option but enters as a wealth fraction, and the prefactor S is a share price, not a wealth.

- $\rho_{\exp}(0) = S \neq 0$, so a zero error is assigned nonzero risk.

The one-standard-deviation loss of the hedge error, $|g|\, S\sigma\sqrt{h}$ (Lemma 1), is linear in $|g|$; by Lemma 2 its interval is empty. The interval in Figure 2 is therefore a consequence of the incorrect functional. Curvature in g has to be introduced through the risk functional.

5 Risk functionals for the hedge error

Let X be the hedge-error P&L over the horizon. By Lemma 1, with the drift excluded, X is Gaussian with mean zero and standard deviation $s(g) = |g|\, S\sigma\sqrt{h}$.

(a) Variance penalty. Evaluate a random P&L X by

$$\mathcal{F}_\gamma[X] = -\frac{1}{\gamma} \log \mathbb{E}[e^{-\gamma X}], \qquad \gamma > 0, \tag{9}$$

the free energy of X at inverse temperature γ (units 1/\$). Negative outcomes are weighted by $e^{-\gamma X}$; $\mathcal{F}_\gamma \to \mathbb{E}X$ as $\gamma \to 0$. The cumulant expansion is

$$\mathcal{F}_\gamma[X] = \kappa_1 - \tfrac{\gamma}{2}\kappa_2 + \tfrac{\gamma^2}{6}\kappa_3 - \ldots, \tag{10}$$

which terminates at $\mathbb{E}X - \frac{\gamma}{2}\operatorname{Var} X$ for Gaussian X. The hedge error therefore contributes

$$\rho_{\text{var}}(g) = \tfrac{1}{2}\gamma S^2\sigma^2 h\, g^2. \tag{11}$$

(b) Constraint on the standard deviation. Impose $s(g) \leq L_{\max}$ on the position. Maximising $\mathcal{F}_\gamma$ under this constraint with Lagrange multiplier $\lambda \geq 0$ is equivalent to maximising $\mathcal{F}_\gamma - \lambda\,(s - L_{\max})$. By complementary slackness $\lambda = 0$ if the constraint is inactive and $\lambda > 0$ if it is active; λ is dimensionless (dollars of objective per dollar of standard deviation). The hedge error contributes

$$\rho_{\text{sd}}(g) = \lambda S\sigma\sqrt{h}\,|g|, \tag{12}$$

which is linear in $|g|$ because the standard deviation is homogeneous of degree one in the position. Hull describes trading desks operating under limits on the Greeks of their positions [1, ch. 19]; a constraint on s is of the same type.

Combined functional.

$$\boxed{\rho(g) = k_1|g| + \tfrac{1}{2}k_2 g^2, \qquad k_1 = \lambda S\sigma\sqrt{h}, \quad k_2 = \gamma S^2\sigma^2 h} \tag{13}$$

with k_1 in \$/share and k_2 in \$/share^2. Both coefficients are built from the one-share standard deviation $S\sigma\sqrt{h}$, to the first and second power respectively. In Lemma 2, $\rho'(0^+) = k_1$ and $\rho'' = k_2$, so the interval is $(k_1,\, k_1 + k_2 u)$ with length $k_2 u$. The linear term makes $x^* = u$ attainable for $b > 0$; the quadratic term gives the interval nonzero length.

6 Solution

With (13),

$$\rho(u) - \rho(u - x) = (k_1 + k_2 u)\, x - \tfrac{1}{2}k_2 x^2, \tag{14}$$

which is concave in x, and for $x > 0$

$$\mathcal{N}(x) = (k_1 + k_2 u - bS)\, x - \tfrac{1}{2}k_2 x^2 - a. \tag{15}$$

Define

$$w = \frac{bS - k_1}{k_2}. \tag{16}$$

Theorem 1 (Optimal correction)**.** *Let $k_1 \geq 0$, $k_2 > 0$. Then*

$$x^* = \begin{cases} u & \textit{(full)} \quad \textit{if } bS \leq k_1 \textit{ and } (k_1 - bS)\,u + \frac{1}{2}k_2u^2 > a, \\ u - w & \textit{(partial)} \quad \textit{if } k_1 < bS < k_1 + k_2u \textit{ and } \frac{1}{2}k_2(u-w)^2 > a, \\ 0 & \textit{(none)} \quad \textit{otherwise.} \end{cases} \tag{17}$$

In the partial region $\mathcal{N}(x^*) = \frac{1}{2}k_2(u-w)^2 - a$, and the residual error is w, independent of u.

Proof. ρ is convex on $[0, u]$ with $\rho'(g) = k_1 + k_2 g$ on $(0, u]$ and $\rho'(0^+) = k_1$; apply Lemma 2 with interval $(k_1,\, k_1 + k_2u)$. *Full:* $\mathcal{N}(u) = (k_1 - bS)u + \frac{1}{2}k_2u^2 - a$. *Partial:* $\rho'(u - \hat{x}) = bS$ gives $u - \hat{x} = w$, with $0 < w < u$ inside the interval. Since $k_1 + k_2u - bS = k_2(u-w)$,

$$\mathcal{N}(\hat{x}) = k_2(u-w)^2 - \tfrac{1}{2}k_2(u-w)^2 - a = \tfrac{1}{2}k_2(u-w)^2 - a.$$

None: case 3 of the lemma, or $\mathcal{N} \leq 0$ at the candidate. □

The fixed cost a enters only the conditions $\mathcal{N} > 0$; it does not affect the value of x^* in the partial region. The residual error after a partial correction is w regardless of the initial error u.

Proposition 1 (Region boundaries)**.** *For fixed u, the boundaries in the (b, a) plane are*

$$\textit{full / partial:} \quad b = b_0 \equiv \frac{k_1}{S}, \qquad 0 \leq a \leq a_c, \tag{18}$$

$$\textit{partial / none:} \quad b = \frac{k_1 + k_2u - \sqrt{2k_2a}}{S}, \qquad 0 \leq a \leq a_c, \tag{19}$$

$$\textit{full / none:} \quad b = \frac{k_1 + \frac{1}{2}k_2u - a/u}{S}, \qquad a \geq a_c, \tag{20}$$

with $a_c = \frac{1}{2}k_2u^2$. The three curves meet at the triple point (b_0, a_c). The partial region is non-empty only for $a < a_c$; at $a = 0$ it is $b \in (b_0,\, b_0 + k_2u/S)$.

Proof. (18) is the boundary $bS = k_1$ of case 1. On it the full and partial candidates coincide ($w = 0$) with $\mathcal{N} = \frac{1}{2}k_2u^2 - a$, so it separates the full and partial regions if and only if $a < a_c$. (19) is the solution of $\frac{1}{2}k_2(u-w)^2 = a$ for b, and (20) the solution of $(k_1 - bS)u + \frac{1}{2}k_2u^2 = a$. At $a = a_c$ both give $b = b_0$. □

Corollary 1 (Width of the partial region; limits)**.** *At $a = 0$ the relative width of the partial region is*

$$\frac{k_2u}{k_1} = \frac{\gamma\,(uS\sigma\sqrt{h})}{\lambda}, \tag{21}$$

a dimensionless ratio: γ times the standard deviation of the P&L of the full error, divided by λ. In the limits:

- *$\gamma \to 0$: the partial region contracts to the line $b = b_0$ and $x^* \in \{0, u\}$;*
- *$\lambda \to 0$: $b_0 \to 0$, the full region is empty for $b > 0$, and every nonzero x^* is partial.*

The partial region is narrow compared with b_0 if and only if $\gamma\, uS\sigma\sqrt{h} \ll \lambda$.

Remark 1 (Dependence on h). Since $k_1 \propto \sqrt{h}$ and $k_2 \propto h$, the relative width scales as $\sqrt{h}$. The boundary $b_0 = \lambda\sigma\sqrt{h}$ is independent of S.

7 Numerical example

Parameters: $S = 49$, $\sigma = 0.20$, $n = 10^5$ options, error 0.40 shares per option ($u = 40{,}000$ shares), $h = 1/52$ year, $a = \$20$, $\lambda = 0.036$, $\gamma = 1.33 \times 10^{-7}$ per dollar. The values of λ and γ are illustrative and are not calibrated. They give

$$k_1 = 0.0489 \text{ \$/share}, \qquad k_2 = 2.46 \times 10^{-7} \text{ \$/share}^2,$$
$$b_0 = 0.100\%, \qquad b_0 + k_2 u/S = 0.120\%, \qquad \frac{k_2 u}{k_1} = 0.20, \qquad a_c = \$197.$$

Table 1: Optimal correction at four values of the proportional cost.

b	$f^* = x^*/n$	Region	$\mathcal{N}(x^*)$ (\$)
0.050%	0.400	full	1,153
0.105%	0.297	partial	88
0.112%	0.158	partial	10
0.130%	0.000	none	0

Figure 3 shows $\mathcal{N}(f)$ for the four values of Table 1. Figure 4 shows $f^*(b)$: $f^* = u/n$ for $b \le b_0$, a decreasing segment in the partial region, and $f^* = 0$ beyond it. The length of the decreasing segment grows with γ, consistent with Corollary 1. Figure 5 shows the regions in the (b, a) plane, evaluated pointwise from (17), with the boundaries (18)–(20) superimposed. A grid maximisation of $\mathcal{N}$ reproduces the closed-form x^* at all values tested.

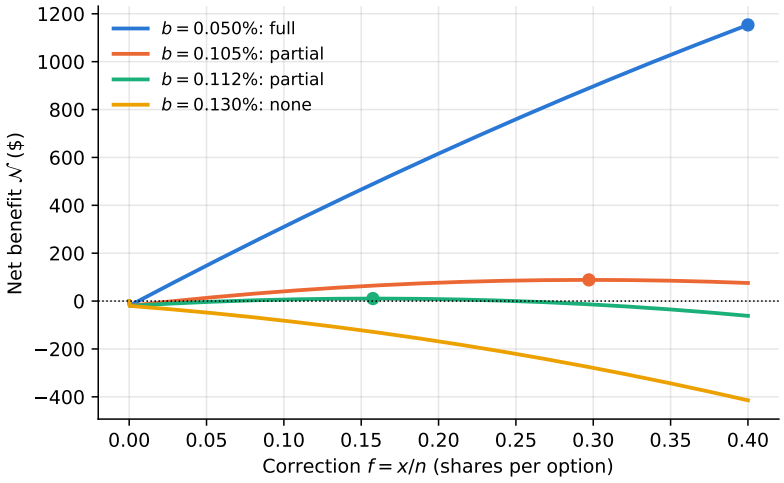


Figure 3: Net benefit $\mathcal{N}$ against the correction per option, for the values of b in Table 1. Markers indicate x^* where $x^* > 0$.

8 Simulation of the complete hedge

Theorem 1 concerns a single decision. To examine repeated application, we simulate the example of [1, ch. 19]: 10^5 short calls with $S_0 = 49$, $K = 50$, $r = 0.05$, $\sigma = 0.2$ and 20 weeks to expiry, sold at the Black–Scholes price (\$240,053 in total). The share price follows geometric Brownian motion with $\mu = 0.13$. The bank account accrues at r and is charged for share purchases and transaction costs. At expiry the shares are valued at S_T and the payoff is paid. Transaction costs are $a = \$20$ and $b = 0.1\%$. Each strategy is evaluated on the same 10^4 paths.

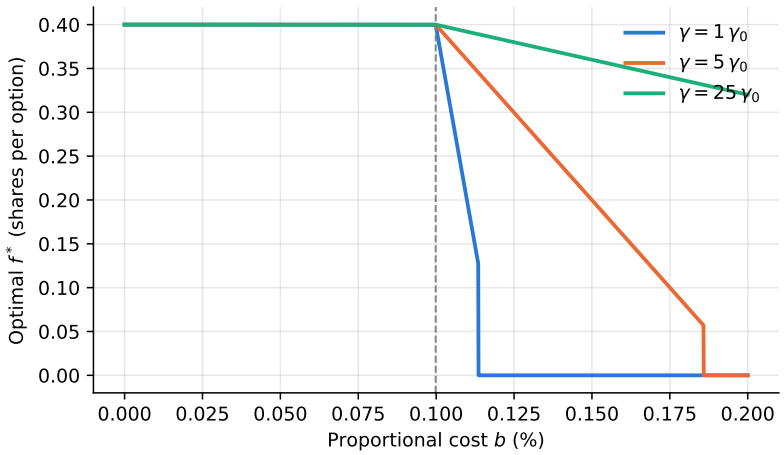


Figure 4: Optimal correction per option against b for three values of γ ($\gamma_0 = 1.33 \times 10^{-7}$). Dashed line: $b = b_0$.

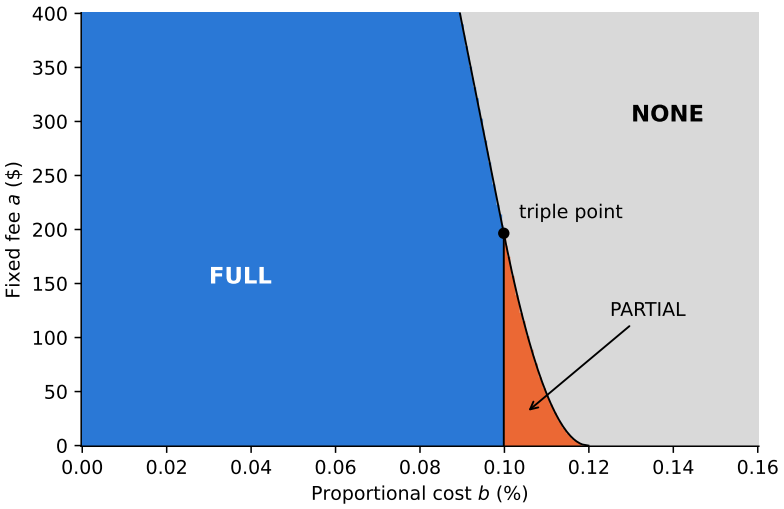


Figure 5: Regions in the (b, a) plane: full (blue), partial (orange), none (grey). Black curves: boundaries (18)–(20), meeting at the triple point (b_0, a_c).

Without transaction costs and with full rebalancing at m equally spaced dates, the mean P&L is consistent with zero and its standard deviation divided by the premium decreases approximately as $m^{-1/2}$ (Table 2), consistent with Hull's results.

Table 2: No transaction costs; full rebalancing at m dates.

m	Mean P&L ($)	Std. dev. / premium
5	−242	0.370
20	−130	0.189
80	−11	0.097

With transaction costs two strategy families are compared:

- full rebalancing at $m \in \{5, 10, 20, 50, 100\}$ equally spaced dates;
- rule (17) evaluated at 100 equally spaced dates, with $h = 1/52$, $\gamma = 10^{-5}$ per dollar and λ a multiple of $\lambda_* = b/(\sigma\sqrt{h}) = 0.036$, the value for which $b = b_0$.

By Theorem 1, for $\lambda < \lambda_*$ every nonzero trade is partial; for $\lambda > \lambda_*$ every nonzero trade is full.

Table 3: Complete hedge: mean number of trades per path, mean transaction cost, and standard deviation of the final P&L (present values, $).

Strategy	λ	Trades	Mean cost	Std. dev.
Full, 5 dates	–	5.0	5,537	89,238
Full, 10 dates	–	10.0	7,275	64,230
Full, 20 dates	–	20.0	9,557	45,928
Full, 50 dates	–	49.9	14,223	29,902
Full, 100 dates	–	99.6	19,789	22,059
Rule, 100 dates	$0\,\lambda_*$	37.0	11,561	26,410
Rule, 100 dates	$0.5\,\lambda_*$	46.6	14,053	23,526
Rule, 100 dates	$0.9\,\lambda_*$	58.4	17,200	22,389
Rule, 100 dates	$1.1\,\lambda_*$	65.9	18,517	22,231
Rule, 100 dates	$2\,\lambda_*$	82.0	19,359	22,110

Results (Table 3, Figure 6):

- $\lambda = 0.5\lambda_*$: mean cost \$14.1k and standard deviation \$23.5k, against \$14.2k and \$29.9k for full rebalancing at 50 dates: equal cost to within 2%, 21% lower standard deviation.
- $\lambda = 0$: \$11.6k and \$26.4k, lower in both quantities than full rebalancing at 50 dates.
- $\lambda = 2\lambda_*$: \$19.4k and \$22.1k, within 3% in cost and 1% in standard deviation of full rebalancing at 100 dates (\$19.8k, \$22.1k). In this case all trades are full and the rule differs from the 100-date schedule only by omitting trades for small errors.

The configurations with partial trades ($\lambda < \lambda_*$) lie below the full-rebalancing curve in Figure 6; those with full trades ($\lambda > \lambda_*$) lie close to its 100-date end point.

9 Limitations

- **Single period.** The functional evaluates the error over one horizon h and omits the cost of later corrections of the residual error. The results depend on h (Remark 1). The multi-period problem is a dynamic-programming problem and is not treated here.

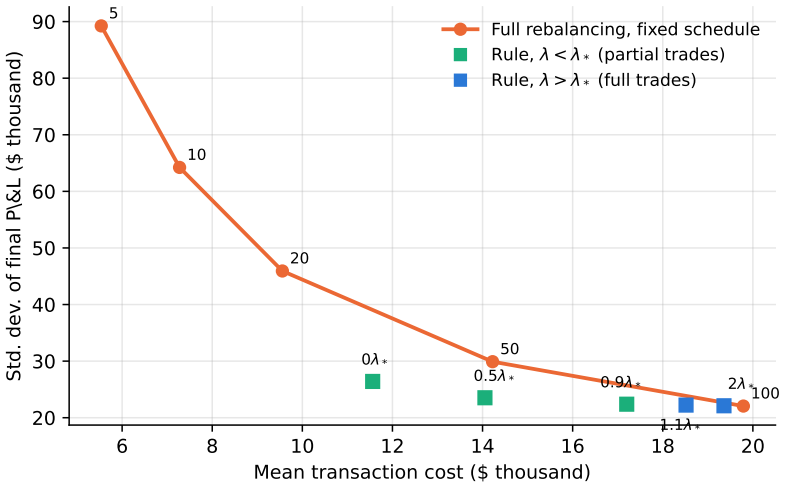


Figure 6: Mean transaction cost against standard deviation of the final P&L. Labels on the full-rebalancing curve give m.

- **Constant error over the horizon.** Changes in Δ during the horizon (gamma) are neglected at leading order.
- **Gaussian approximation.** $\mathcal{F}_\gamma = \mathbb{E}X - \frac{\gamma}{2}\operatorname{Var} X$ holds exactly only for Gaussian X; non-zero third and higher cumulants of an option position contribute at higher order in γ.
- **Parameters.** λ and γ are illustrative. A calibration, for example to a target standard deviation of the final P&L, is not performed.
- **Cost model.** Market impact, size-dependent proportional costs and the cost of closing the position at expiry are not included.

10 Summary

To leading order in the horizon, the P&L of a delta-hedge error is linear in the error, so the dependence of the optimal correction on the error size is determined by the risk functional. For convex ρ, a partial correction is optimal only if bS lies in $(\rho'(0^+), \rho'(u))$ (Lemma 2). A functional linear in the P&L standard deviation gives an empty interval; a variance penalty gives an interval with lower endpoint zero. For the combined functional $\rho = k_1|g| + \frac{1}{2}k_2 g^2$ the optimal correction is given in closed form, with three regions in the (b, a) plane, explicit boundaries and a triple point at $(k_1/S, \frac{1}{2}k_2u^2)$. The partial region has relative width $\gamma u S\sigma\sqrt{h}/\lambda$. An exponential functional that produced partial optima only near $b = 24\%$ was shown not to represent the risk of the hedge error. In the simulated 20-week hedge, configurations of the rule with partial trades attain lower P&L standard deviation than full rebalancing on a fixed schedule at equal mean cost.

A Implementation

All figures and tables are produced by `make_figures.py`. Implementation of (17):

```
def decide(u, S, b, a, k1, k2):
    """Optimal number of shares to trade for a hedge error u >= 0 (Theorem 1)."""
    c = b * S                                  # marginal cost per share
    if c <= k1:                                # marginal cost <= rho'(0+): full candidate
```

```
    x = u
elif k2 > 0 and c < k1 + k2 * u:       # rho'(0+) < c < rho'(u): interior candidate
    x = u - (c - k1) / k2
else:                                  # c >= rho'(u): no trade
    return 0.0
gain = (k1 + k2 * u) * x - 0.5 * k2 * x**2 - (a + c * x)
return x if gain > 0 else 0.0
```

References

[1] J. C. Hull, *Options, Futures, and Other Derivatives*, 11th ed., Pearson, 2021.

[2] A. E. Whalley and P. Wilmott, An asymptotic analysis of an optimal hedging model for option pricing with transaction costs, *Mathematical Finance* **7**(3), 307–324, 1997.